



Definition

- Study of Cell - Cytology
- Building blocks of life
- Fundamental unit of life
- Structural or functional basis of life

Discovery

- 1st by Robert Hooke (1665) - in cork ((Dead Cell)) - Father of cytology
- by Antonie van Leeuwenhoek (1674) - in pond water (Living Cell)
- by Robert Brown (1831) - Nucleus
- by Fredrick Miescher (1869) - Molecule discovery of DNA
- Watson & Crick (1953) - Double-helical structure of DNA
- by Camillo Golgi - Golgi bodies

Cell Theory

- All the plants & animals are composed of cells
- By Schleiden & Schwann (1838)
- Omnis Cellula E Cellula - by Rudolf Virchow - New cells are formed from the pre-existing cells

Cell cycle

Types of cell

Prokaryotic Cell	Eukaryotic Cell
- Size: 1-10 um - Nucleus absent - Nucleolus absent - Cell organelles absent - Chromosome: single Example: Bacteria, Cyanobacteria (Blue green algae), mycoplasma	- Size: 5-10 um - Nucleus present - Nucleolus present - Cell organelles present - Chromosome: more than one Example: Plant and Animal cell

Cell membrane

- Outermost covering of the cell
- Thin, elastic, flexible, Living
- provide definite shape & holds content
- Selectively permeable
- Proteins + lipids (Composition)
- Diffusion - Spontaneous movement of molecules from high to lower concentration
- Osmosis - Movement from high to low but through a semipermeable membrane
 - Hypertonic - Solute > Solvent
 - Hypotonic - Solute < Solvent
 - Isotonic - Solute = solvent

Cell wall

- Provides shape, strength & turgidity
- Freely permeable, Non-living
- Plants - made up of Cellulose
- Bacterial - made up of Peptidoglycan/Peptin
- Fungi - made up of Chitin

Nucleus

- "Brain of the cell"
- Nuclear envelope - Composed of membranes & Covers Nucleus - Separates nucleus from cytoplasm & permits materials between them - Liquid ground substance of nuclear envelope
- Nucleoplasm - one or more in number - Ribosomes - forms protein forming centres
- Nucleolus - Entangled mass of thread like structures - During cell division they get organised in chromosome
- Chromatin material

Chromosome

- Thread like structures & visible during cell division
- Components - DNA - Consists of genes - Proteins - histone
- 23 Pairs - 22 pairs (autosomal) - 1 pair (sex pair - XX & XY)
- Parts between plasma membrane & nuclear envelope
- Cell organelles present in it
- Cytosol - Aqueous ground substance
- Cytoplasm Inclusions - Non-living component in cytoplasm eg. Pigments, nutrients etc

Cytoplasm

Cytoskeleton

- Helps in providing structural support & movement to cell
- Centrosome - Made of two Centrioles in cylindrical & 9 + 0 / 9 + 2 Form - Microtubules organising centre
- DNA Replication - cell copies its DNA before dividing, ensuring each new cell has identical copy of original DNA

4 base pairs (in DNA)	Base pairs (in RNA)
Adenine - Thymine	Adenine - Uracil
Guanine - Cytosine	Guanine - Cytosine

Enzymes essential for DNA replication in an animal cell

- Topoisomerase: Prevents DNA from getting tangled (cavoids supercoiling)
- DNA Ligase: This enzyme facilitates the joining of DNA strands together
- DNA Polymerase: This enzyme doubles the amount of DNA in a cell during cell division (builder)
- DNA primase: It initiates DNA replication (new strands are produced)
- Helicase: Unwinding enzyme

Cell Division

- Mitosis - occurs in Somatic cells - Nucleus divides once - diploid (2), 2 daughter cells
- Meiosis - occurs in Germ Cells - Nucleus divides twice - haploid (1), 4 daughter cells

Cell organelles

- Mitochondria - Powerhouse of the cell - Sausage shape - Energy currency - ATP (Adenosine Triphosphate) - Outer - Porous - Two-membranes - Inner - ATP synthesis - Has its own DNA & Ribosomes - involved in Apoptosis - suicidal bag of the cell - Filled with Digestive enzyme - breakdown long chain of Fatty acids through beta oxidation - Neutralises harmful toxins - Cis face - towards nucleus (Convex) - Trans face - towards plasma membrane (Concave)
- Lysosomes - involved in Apoptosis - suicidal bag of the cell
- Peroxisome - breakdown long chain of Fatty acids through beta oxidation - Neutralises harmful toxins
- Golgi bodies - Cis face - towards nucleus (Convex) - Trans face - towards plasma membrane (Concave) - Packaging & Dispatching - Stores, modification of package - Forms lysosomes - Synthesis of proteins, digestive enzyme - ribosomes present
- Rough ER - synthesis of fats & lipids, Detoxify poisons & Drugs - ribosomes absent
- Smooth ER - Long Tubules, Vesicles & Cisternae - Large, distinct, Permanent - Provide turgidity & rigidity - Store cell sap (consist of - Amino acids, Sugars, Organic acid)
- Vacuoles - 50-90% of cell - Small, temporary - stores glucose, glycogen, protein

Plastids

- Only in plant cell
- Has its own DNA & Ribosomes
- Chloroplasts - stores Chlorophyll pigment
- Chromoplasts - stores colour pigment
- Amyloplast - Storage of Carbohydrates
- Leucoplasts - stores glucose, glycogen, protein